Nihar Gaddam

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Computer Engineering – Applied Advanced Study — *Dec 2023*

- GPA: 3.75/4.00
- Relevant Coursework: Distributed Systems, Cloud Computing, Foundations of Computer Systems, Computer Vision, Machine Learning for Signal Processing, Introduction to Deep Learning

Sreenidhi Institute of Science and Technology — Hyderabad, India
Bachelor of Technology in Electronics and Communication Engineering — *June 2020*

- GPA: 9.39/10.00 • Minor in Computer Science and Engineering

Skills

Languages: Go, Rust, Python, C++, Java
IAAS: AWS, Azure, GCP
Container Platform: Docker, Kubernetes
CI/CD: GitHub Actions, Apache Airflow
IAC & Config Management: Terraform, Puppet
Storage: HBase, HDFS, Zookeeper, MySQL, MongoDB, Neo4j, Postgres
Frameworks/Tools: PyTorch, Spark, Kafka, Vertex AI
Monitoring and Observability: Prometheus, Grafana, Splunk, OpenTelemetry, SignalFx

Experience

Cisco Systems Inc. — Pittsburgh, PA
Software Development Engineer — *Aug 2024 – Present*

Backend Engineering & API Development

- Designed and maintained features for a Go-based Kubernetes operator (control plane) managing the full lifecycle of thousands of hosted Splunk Cloud deployments — the isolated, multi-tenant product instances used by customers and internal teams — plus a customer-facing configuration API.

System Monitoring & Performance Optimization

- Held 24/7 on-call across 5 Kubernetes clusters and 8 services plus the control plane, triaging and resolving production incidents on page while reducing technical debt through targeted refactors across development, staging, and production.

Key Projects and Achievements

- Delivered a declarative, self-serve workflow to replace months-long manual provisioning of preview deployments (short-lived demo instances teams trial before adopting the official version), with validations that strictly isolate preview from production; scaled to ~100/month with zero manual coordination and drove adoption across five teams.
- Built the platform's first metrics + SLO observability layer (Prometheus, OpenTelemetry), instrumenting per-phase latency and success/failure metrics across every provisioning stage — infrastructure, software, and final delivery — with data-driven SLOs; shipped pageable alerts that catch genuine regressions proactively and cut cross-team mispages by 21%.
- Diagnosed and fixed a disaster-recovery teardown ordering bug that could destroy a live primary deployment while its replica was still active; redesigned the control loop to enforce dependency-safe teardown, preventing data loss and eliminating the manual-cleanup incident tickets and on-call pages those mis-ordered teardowns had generated.
- Built an automated garbage-collection system for stale infrastructure, reclaiming 2,385 abandoned deployments and cutting unnecessary cloud spend fleet-wide in production alone.
- Automated service-to-service authentication with ephemeral credentials, removing the need to raise manual tickets to update the latest build version, and drove encryption-at-rest to 100% coverage across 3,000+ deployments.
- Closed a network security-group filtering gap, preventing all related incidents, and surfaced certificate-rotation status to customers, cutting support tickets 23% for 8,000+ users.
- Authored a design proposal integrating a downstream pooling service with the control plane to pre-provision Azure subscriptions for deployment networking (otherwise created on demand), raising Azure network provisioning success from 33% to 98% and cutting provisioning time to ~20 min.
- Automated authentication-token rotation across 15 service accounts, eliminating manual intervention and improving token-management efficiency ~33%.

Accenture Solutions Pvt. Ltd. — Hyderabad, India
Application Development Associate — *Dec 2020 – Oct 2021*

Cloud Infrastructure Management

- Migrated client infrastructure from on-premises to AWS and Azure, improving scalability and reducing operational costs by 20%; maintained development and production environments, reducing downtime by up to 40%.
- Executed high-risk production changes with minimal disruption, maintaining a 99.8% success rate in change management.

Data Analysis and Reporting

- Leveraged Power BI for real-time server performance trend analysis, reducing response time to performance issues by 30%; managed ETL dataflows from Azure SQL, SharePoint, Redshift, MongoDB, and Snowflake into BI reporting pipelines.

Projects

Personal

timberdb | *Go* Jun 2026

- Built a time-partitioned, TTL-native LSM storage engine in Go; benchmarked 2.0x Pebble, 5.8x Badger, and 13x bbolt append throughput at 3 allocs/op, with 6.6x faster scans than Badger.

SVID-Exchange | *Go* Mar 2026

- Built a Zero Trust token exchange service that converts SPIFFE SVIDs into scoped, short-lived ES256 JWTs over mTLS, eliminating static shared secrets for service-to-service authentication.

Saga-Conductor | *Go* Mar 2026

- Built a lightweight saga-pattern orchestrator for distributed transactions: callers define each step with a forward action and a compensating action, and the conductor executes them in order, automatically rolling back completed steps in reverse on failure.

Academic

Distributed Backend Storage for Multiplayer Environments | *Go* Dec 2023

- Created a distributed key-value storage system utilizing replication, partitioning, and concurrency mechanisms to support a globally distributed multiplayer online game, ensuring data consistency and low-latency access across multiple backend servers.

Analysis of Twitter User Graph with Spark | *Scala* Nov 2023

- Proficiently applied Apache Spark programs on Azure HDInsight to analyze the Twitter social graph; implemented the PageRank algorithm to identify influential users and optimized runtime from 60 to 28 minutes, enhancing performance.

Advanced Resource Scaling using Cloud SDKs | *Python* Sep 2023

- Implemented horizontal scaling with the Boto3 AWS SDK, reducing response times from 30 to 8 minutes; increased throughput from 8K to 40K RPS; configured Elastic Load Balancers, Auto Scaling Groups, and Terraform for efficient deployment.

Certifications

Certified Kubernetes Application Developer (CKAD) — Cloud Native Computing Foundation (CNCF) 2024